THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM ONE ANNUAL EXAMINATION
031 PHYSICS
MARLING SCHEME NOVEMBER 2025

SECTION A (15 MARKS)

1. Answers 10marks 01mark @

i.	ii.	iii.	iv.	v.	vi.	vii.	viii.	ix.	x.
B	B	B	D	C	B	B	C	A	C

2. Answers total 05 marks 1@

i.	ii.	iii.	iv.	v.
C	D	F	H	G

SECTION B (70 MARKS)

3.

Physical quantity	Instrument used	SI unit
i. Length 0.5marks	Ruler 0.5marks	Metre 01 marks
ii. Mass 0.5marks	beam balance0.5marks	Kilograms 01 marks
iii. time 0.5marks	stopwatch0.5marks	Seconds 01 marks
iv. temperature0.5marks	Thermometer 0.5marks	Degree Celsius (°C) or Kelvin (K). 01 marks
v. volume 0.5marks	Measuring cylinder0.5marks	cubic metre (m^3)01 marks

4. (a)
   - Average density of the body must be less than density of fluid…..01 marks
   - The upthrust due to fluid must be greater or equal to the total weight of the body /object 01 marks
   - The volume of the submerged part of an object must be large enough to displace a large volume of water 01 marks

(b) Calculate the density of a body if a floating body is 95% submerged in water . The density of water is $1000\frac{kg}{m^3}$ **(03marks)**

Percentage submerged= $\frac{density\ of\ substance}{density\ of\ water} \times 100\%$ ... ...01 marks

95%= $\frac{density\ of\ substance}{1000kg/m^3} \times 100\%$ ... ...01 marks

$\therefore$ $density\ of\ substance = 950kg/m^3$ ... ...01 marks

(c)In an experiment to determine the relative density of the liquid , a solid Q weighed as follows weight in air W_0= 8.6 N , weight in water W_1= 6.0 N, weight in Liquid W_2= 5.4 N determine the relative density of the liquid (04marks)

Given

weight in air = 8.6 N,
weight in water = 6.0 N,
weight in liquid = 5.4 N.

Upthrust in water = weight in air − apparent weight in water = 8.6 − 6.0
= 2.6 N. .(0.5marks)
Upthrust in the liquid = 8.6 − 5.4
= 3.2 N.....(0.5marks)
Relative density of the liquid = upthrust in the liquid /upthrust in water.....
.(01marks)
$=\frac{3.2N}{2.6N} = 1.2$.....(0.5marks)
Relative density of the liquid = 1.2 .(0.5marks)

5. (a) Matter is made of a large number of tiny particles (atoms/molecules) which are in constant random motion. The temperature of a substance is related to the average kinetic energy of these particles.

(b) List three factors that affect the surface tension of given liquid (03marks)

Three factors that affect surface tension (3 marks)

- Temperature — surface tension decreases with increasing temperature. ... ...01 marks
- Presence of impurities or surfactants (soap, detergents) — lowers surface tension. ... ...01 marks
- Nature of the liquid (different liquids have different intermolecular forces → different surface) ... ...01 marks

(c) Give out five applications of capillarity in daily life **(**05marks)

- Water transport in plants (xylem) from roots to leaves.
- Wicking action in cloths and paper towels (absorption).

- Ink flow in fountain pens and capillary tubes in pens.
- Soil moisture movement (capillary rise in soils).
- Thin-layer chromatography (capillary action moves solvent up the plate).
  ... ...01 marks @ application

6. (a)

Compute face areas:

- Face (5 × 10) = 50 cm² ... ...0.5 marks
- Face (5 × 4) = 20 cm² ← smallest area ... ...0.5 marks
- Face (10 × 4) = 40 cm² ... ...0.5 marks

Largest pressure is with the **smallest contact area** → the face **5 cm × 4 cm** (area = 20 cm²) exerts the largest pressure. ... ...01 marks
Reason: Pressure = Force / Area; smaller area → larger pressure for same force.
... ...01.5 marks

(b) State and explain the principle by which a hydraulic press operates, and mention one of its applications **(04 marks)**
Principle: Pascal's law — pressure applied to a confined fluid is transmitted undiminished in all directions. In a hydraulic system a small force on a small-area piston produces the same pressure on a large-area piston, producing a larger output force. ... ...03 marks
Application: Car hydraulic jack, hydraulic brake systems, hydraulic press for metal forming. ... ...01 marks any application of hydraulic press
(c) Explain two factor which affects pressure in liquids **(02marks)**
The pressure in liquid is affected by
Density of the liquid
Height of the liquid column
Acceleration due to gravity any two factor ... ...01 marks@ factor

7. (a)

- **Kinetic energy (KE):** energy due to motion, KE $= \frac{1}{2}mv^2$. ... ...01 marks
- **Potential energy (PE):** energy stored due to position in a force field (e.g., gravitational PE ... ...01 marks

(b) What is the kinetic energy of a 12g bullet travelling at speed of 320m/s **(03marks)**
Mass m = 12 g = 0.012 kg;
speed v = 320 m/s**.**
KE=1/2mv^2 ... ...01 marks
=0.5×0.012×320^2
=0.006×102400 ... ...01 marks

=614.4 J... ... ...01 marks

(c) Mention five forms of energy you have learned about

Kinetic energy.
Gravitational potential energy.
Thermal (heat) energy.
Chemical energy.
Electrical energy. nuclear, light (radiant), elastic potential energy, sound energy heat energy. Any five forms of energy ... ...01 marks@

8. (a) (i) Distance is the length of the path taken by an object while in motion .It has no specific direction while Displacement is the distance covered in a specific direction .....(02marks)

(ii) Speed is the rate of change of distance wile velocity is a rate of change of displacement .....(02marks)

(a) (i) given that a=4 $\frac{m}{s^2}$ time t=5s

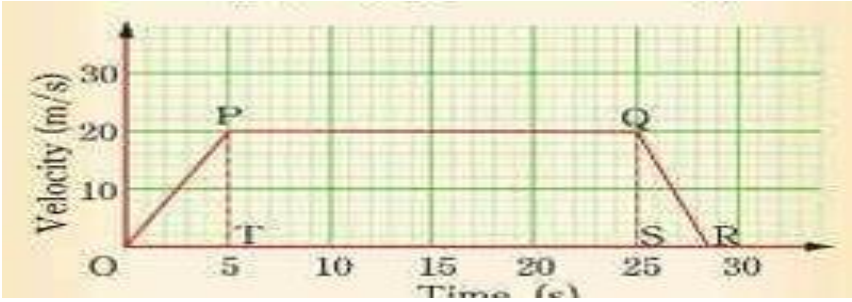
$v = u + at$ .....(0.5marks)

= 0m/s +4 $\frac{m}{s^2}$ x5s .....(0.5marks)

= 20m/s .....(0.5marks)

The maximum speed reached was = 20m/s.....(0.5marks)

(ii) by graphical method the figure below shows the graph of velocity m/s against time (s) (2marks)



The area under the graph represents the total distance covered

= $\frac{1}{2}(PQ + OR) \times PS$.....(0.5marks)

= $\frac{1}{2}(20s + 28s) \times 20m/s$.....(0.5marks)

= 480m.....(01marks)

9. (a) (i) from $t = \frac{\Delta v}{a}$, v= 36km/h= 10m/s a = 2m/s^2.....(01marks)

t=10m/s/(2m/s^2) .....(01marks)

= 5s ……(01marks)

(ii) final velocity = 0m/s because the body was brought to rest ……(02marks)

(b) From $V^2 = U^2 - 2ah$ ……(01marks)

V= om/s , h=20m, g=10m/s^2

02=u^2-2x10x20……(01marks)

U2=400m^2/s^2……(01marks)

U=$\sqrt{400}$……(01marks)

U= 20m/s^2……(01marks)

SECTION C (15 MARKS)

10. (a)

(i) The student likely thought that mass and weight are the same and that both decrease on the Moon. In fact, mass is an intrinsic amount of matter and does not change with location; weight is the gravitational force on that mass and does change with local gravity.

(ii)

- **Mass**: amount of matter in a body. **Same** on Earth and Moon. Unit: kg.
- **Weight**: gravitational force on the mass = $W = mg$. It depends on the local gravitational field

(b) Describe five practical applications of density or relative density in everyday life. (05marks)

- Ship and boat design — ensure average density less than water so floats.
- Hydrometers — determine sugar content or alcohol by measuring relative density.
- Separating oil and water — oil floats because less dense; used in waste-water treatment.
- Determining purity of metals — compare measured density with standard to detect impurities.
- Hot-air balloon operation / buoyancy calculations — density differences between hot and cold air determine lift.

(c)In an experiment to determine the density of liquid X, a Form One student recorded the following results: Mass of beaker = 600 g and Mass of beaker + liquid (25 cm^3) = 700 g. What was the student's calculated density for liquid X?

Data given

Mass of the beaker = 600g

Mass of beaker + mas of liquid= 700g

Volume of the liquid= volume if the beaker = 25cm^3……..01marks

Required density of the liquid

From density= mass/ volume ……..01marks

But mass of the liquid = 700g-600g

= 100g……..01marks

Density = 100g/ 25cm^3 ……..01marks

Density = 4g/cm^3 ……..01marks